Decoding and Steering Graded Causal Implication in Pretrained Language Models

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Abstract

We study whether pretrained language models behaviorally register graded causal implication, whether that information is linearly accessible, whether representational evidence tracks graded judgments, and whether predictable decision failures can be reduced by activation steering. Across seven models, critical items produce both greater surprisal for a subsequent claim of causal ignorance and a larger factor-probability advantage after *because* than after an explicit happenstance continuation. Direct Yes/No and Likert judgments are more mixed: two Qwen models are highly discriminative, while several other models exhibit weak thresholded accuracy and strong order sensitivity. Across six models, causal probes and contrastive activation addition (CAA) directions discriminate held-out causal items well, whereas directions trained on temporal contrasts transfer inconsistently. Representational scores correlate strongly with an external Gemini rating but, for several models, only weakly with the model's own prompted judgments. Development-selected raw activation interventions improve held-out balanced accuracy in several settings. These results distinguish behavioral sensitivity, linear accessibility, and final decision readout.

1 Introduction

When one event is described immediately after another, comprehenders often infer more than temporal succession: the first event may be understood as explaining why the second occurred. This graded causal implication supports ordinary narrative understanding, explanation, and commonsense reasoning. A language model that is sensitive to the distinction should separate genuinely explanatory factors from details that are merely compatible with an event. Yet a model's prompted answer need not transparently reveal whether it represents that distinction. Answer-token priors, scale presentation, and late-stage readout can all distort an otherwise informative internal signal.

This representation–behavior distinction parallels recent work on event plausibility. Lepori et al. show that linear directions in language-model activations can distinguish modal categories and align with fine-grained human judgments even when output probabilities provide an unreliable behavioral measure [Lepori et al., 2026]. Causal implication presents a related but more relational problem: the model must evaluate whether a stated factor explains a particular event, rather than assess the plausibility of one sentence in isolation. Temporal proximity is a particularly important alternative explanation because causes normally precede their effects, but temporal order alone does not establish an explanatory relation.

Before asking how causal implication is represented, we first test whether it is visible in unconstrained next-token behavior. Following the ignorance-continuation logic of Pan and Kehler [2025], an inferred explanation should make a subsequent claim that the reason is unknown less expected. A complementary diagnostic compares the probability of the same factor after *because* versus after the

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explicitly incidental continuation *and it so happened that*. Together, these tests avoid requiring the model to interpret a metalinguistic Yes/No question or numeric scale.

We therefore analyze causal implication at four levels. First, we compare implicit eliciture signatures from the ignorance and explicit-happenstance diagnostics with direct probabilistic Yes/No and Likert judgments. Second, we compare prompted behavior with linear probes and paired mean-difference directions, and test whether causal and temporal representations transfer across domains. Third, we ask whether the resulting continuous representation scores track probabilistic Yes/No judgments, probabilistic Likert judgments, and an external graded rating. Fourth, we distinguish two probe-relative decision failures: *suppression*, in which the causal probe decodes a factor as explanatory but the model rejects it, and *post-hoc attribution*, in which the probe decodes a factor as non-explanatory but the model accepts it. We test whether these failures are predictable from pre-answer activations and whether subtracting development-selected failure directions changes the final decision.

Our experiments are organized around the following research questions:

RQ1: Behavioral eliciture diagnostics. Do causal critical items, relative to matched controls, increase the surprisal of a subsequent ignorance claim and increase the probability of the explanatory factor after *because* relative to *and it so happened that*? How consistently does this implicit sensitivity appear in direct probabilistic Yes/No and Likert judgments?

RQ2: Decodability and transfer. Is graded causal implication linearly decodable across pretrained model families, and is the decoded causal signal distinct from a direction learned for temporal relations? How does internal decodability compare with prompted Yes/No and Likert behavior?

RQ3: Correlation with graded judgments. Do raw causal probe logits and CAA projections track item-level graded judgments from the target model and an external rater, both across individual items and in critical-minus-control pair differences?

RQ4: Predicting and reducing failure modes. Can probe-relative suppression and post-hoc attribution be predicted from pre-answer activations and reduced by format-balanced, development-selected residual-stream steering without substantially damaging probe–output agreements?

2 Methods

2.1 Models

The RQ1 continuation diagnostics evaluate seven pretrained, non-instruction-tuned autoregressive language models: Gemma 2 2B and 9B, Qwen3 4B and 8B, Llama 3 8B, Llama 3.2 3B, and OLMo 3 7B. The direct prompted judgments and RQ2–RQ4 use the same set except OLMo 3 7B. For representation analyses, we use the residual stream at the final prompt token. Computation uses bfloat16 model weights; extracted activations and fitted analysis objects are stored with fixed dataset and checkpoint hashes.

2.2 Data

The main causal training and development sets contain 2,000 rows (1,000 critical–control pairs) and 400 rows (200 pairs), respectively. The held-out causal evaluation set contains 300 rows (150 pairs). Within a pair, the critical item supports the target event as a causal consequence of the stated factor, whereas the control preserves the event and discourse structure while adding a non-explanatory factor. The revised training and development files are `rc_modifier_train_set.txt` and `rc_modifier_dev_set.txt`; these historical filenames contain the revised independent-clause stimuli used in this analysis. Held-out items are distinct from training and development items.

For the temporal ablation, we use 800 training, 200 development, and 200 test pairs. Positive and negative temporal items contrast an event-consistent temporal relation with a matched non-temporal control. The causal and temporal data are fit independently; cross-domain evaluations apply a frozen probe or direction from one domain to the other.

2.3 RQ1: behavioral eliciture diagnostics

RQ1 uses the 300 held-out causal rows (150 matched critical–control pairs), equally divided between relative-clause and modifier constructions. Scores use raw, untempered next-token log probabilities. For the ignorance diagnostic [Pan and Kehler, 2025], we remove the context's terminal period and append the exact continuation "`, and I don't know why.`" The row score is the negative summed log probability of all continuation tokens,

$$I_i = -\log P(\textit{and I don't know why} \mid x_i), \tag{1}$$

so a positive critical-minus-control gap indicates that professing ignorance is less compatible with the inferred explanation in the critical item.

For the explicit-happenstance diagnostic, let E_i be the event and F_i the factor. Let B_i denote the reconstruction prefix ending in E_i followed by *because*, and let S_i denote the matched prefix ending in E_i followed by *and it so happened that*. We score the identical factor tokens after these prefixes and compute

$$H_i = \overline{\log P}(F_i \mid B_i) - \overline{\log P}(F_i \mid S_i). \tag{2}$$

The mean over factor tokens prevents target length from determining the score. A positive critical-minus-control gap means that causal framing provides a larger relative advantage for explanatory than irrelevant factors. RQ1 does not use the separate *because*–versus–bare-*and* diagnostic.

For each measure, we report critical and control means, the matched critical-minus-control gap, a 95% interval from 10,000 pair-bootstrap samples, pairwise accuracy with ties worth one half, row-level ROC-AUC, and a one-sided paired sign-flip test. No threshold or prompt is selected on the test set. The explicit-happenstance contrast was prespecified as a secondary control in the original behavioral protocol, so its interval and unadjusted sign-flip test are interpreted separately from that protocol's primary FDR family.

2.4 RQ1: prompted behavioral measurements

For Yes/No judgments, we score pooled log-probability mass over capitalization and leading-space variants of *Yes* and *No* and normalize within the two answer classes,

$$p_{\text{yes}} = \frac{P(Y)}{P(Y) + P(N)}. \tag{3}$$

We use both "Answer Yes or No" and "Answer No or Yes." For Likert judgments, we score all integer candidates $k \in \{1, \ldots, 7\}$ and use their posterior expectation,

$$r = \sum_{k=1}^{7} k \frac{\exp q_k}{\sum_{j=1}^{7} \exp q_j}, \qquad \tilde{r} = \frac{r-1}{6}. \tag{4}$$

We use both ascending (1-to-7) and reversed (7-to-1) anchor descriptions. A score strictly above 0.5 is classified as positive; ties are classified as negative. We report each presentation order separately and an item-level average across the two orders. The external ceiling is one frozen 1–7 Gemini 3.1 Pro Preview rating per unchanged test item, normalized as $(r-1)/6$.

2.5 RQ2: probes, CAA, and transfer

For each candidate layer ℓ, we fit a class-balanced logistic regression probe to distinguish critical from control items. The probe score is its raw decision logit. Regularization and the layer are selected on the corresponding development set, with the earliest layer breaking ties.

The CAA direction is the mean paired activation difference,

$$v_\ell = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \left(h_{i,\ell}^{\text{critical}} - h_{i,\ell}^{\text{control}} \right). \tag{5}$$

An item is scored by its projection onto the unit direction, $s_{i,\ell} = h_{i,\ell}^\top v_\ell / \|v_\ell\|_2$. CAA layers maximize the mean of development ROC-AUC and paired accuracy. Probe and CAA thresholds are chosen on the source-domain development split by balanced accuracy and then frozen. We report ROC-AUC, balanced accuracy, and paired accuracy on causal and temporal tests, including causal-to-temporal and temporal-to-causal transfer.

2.6 RQ3: correlation with graded judgments

We correlate the frozen causal probe logits and unit-direction CAA projections with (i) order-specific and order-averaged Yes/No probabilities, (ii) probabilistic Likert scores, and (iii) frozen Gemini ratings. Primary analyses use row-level Pearson and Spearman correlations. Secondary analyses first compute the critical-minus-control difference for each pair and correlate those differences. We keep representational scores raw: Pearson correlation is unchanged by positive affine rescaling, and Spearman correlation depends only on ranks. Ninety-five percent confidence intervals use 1,000 bootstrap resamples clustered by pair ID.

2.7 RQ4: predicting failure modes

Let $c \in \{0, 1\}$ be the decision of the previously trained, frozen causal probe, using a raw-logit threshold of zero, and let $b \in \{0, 1\}$ be the model's prompted behavioral decision. Each item–prompt observation receives one of three labels. *Suppression* has $c = 1, b = 0$; *post-hoc attribution* has $c = 0, b = 1$; and probe–output agreements ($b = c$) are *correct*. The terms *correct positive* and *correct negative* therefore mean positive and negative agreement with the causal probe, not necessarily agreement with dataset ground truth. Appendix A visualizes the four combinations. Prompt-order variants of the same item may receive different failure labels.

At every layer, a class-balanced multinomial logistic regression receives the standardized final-token residual activation concatenated with a one-hot prompt-variant indicator. It is trained on all four formats in the semantically independent `rc_modifier_train_set2.txt`. We select the layer on `rc_modifier_dev_set2.txt` by macro-F1, then balanced accuracy, with the earliest layer as the final tie-breaker. The selected probe is frozen for the 1,200 held-out observations (300 items under four formats). We emphasize balanced accuracy and macro-F1 because the correct class is often much larger than either error class.

2.8 RQ4: vector construction and intervention

We construct separate endpoint-balanced vectors for Yes/No and Likert. For suppression at layer ℓ,

$$\begin{aligned} v_{\mathrm{sup},\ell} = \mathbb{E}[h_\ell \mid \text{suppression}] \\ - \mathbb{E}[h_\ell \mid \text{correct positive}], \end{aligned} \tag{6}$$

For post-hoc attribution, the matched contrast reverses the gold label but retains the same failure-minus-correct orientation:

$$\begin{aligned} v_{\mathrm{post},\ell} = \mathbb{E}[h_\ell \mid \text{post-hoc attribution}] \\ - \mathbb{E}[h_\ell \mid \text{correct negative}]. \end{aligned} \tag{7}$$

Contrasts are computed within each prompt order and then averaged, giving the two orders equal weight. Vectors are raw and are not length-normalized.

Vector layers maximize development AUC for separating the target failure from its matched-correct class. For $\alpha \in \{0, 0.5, 1, 2, 3\}$, we intervene at the selected layer with

$$h'_\ell = h_\ell - \alpha v_{f,\ell}. \tag{8}$$

Magnitude is selected on development data by balanced accuracy minus the rate at which initially correct examples become incorrect. At test time, the primary intervention is gated by the frozen failure probe; oracle gating is retained as a diagnostic. We report balanced accuracy, damage rate, net corrections, and the signed change in the Yes–No or Likert margin.

3 Results

3.1 RQ1: continuation likelihoods are robust, but direct judgments are mixed

Every model shows a positive ignorance gap (Table 1). Critical items increase continuation surprisal by 1.856–3.165 summed nats relative to matched controls. Pairwise accuracy ranges from 0.873 to 0.933, and row-level ROC-AUC ranges from 0.722 to 0.821. Thus, after a context supports an

	IDK			Because vs. And		
Model	Logprob Diff ± 95% CI	Pair acc.	AUC	Logprob Diff ± 95% CI	Pair acc.	AUC
Gemma 2 2B	2.473 ± .358	.887	.737	.454 ± .063	.940	.894
Gemma 2 9B	3.165 ± .386	.927	.782	.411 ± .054	.927	.875
Qwen3 4B	2.371 ± .334	.873	.758	.403 ± .063	.867	.845
Qwen3 8B	2.914 ± .373	.893	.821	.280 ± .063	.760	.750
Llama 3.2 3B	1.856 ± .287	.887	.722	.361 ± .053	.893	.860
Llama 3 8B	2.997 ± .301	.927	.817	.399 ± .047	.907	.907

Table 1: Behavioral eliciture diagnostics on 150 matched critical–control pairs. Log-probability differences are mean within-pair critical-minus-control differences; CI values are symmetric half-widths of the 95% pair-bootstrap intervals. Ignorance uses summed continuation surprisal; explicit happenstance uses the mean factor-token log-probability advantage after *because* versus *and it so happened that*.

	Yes/No order average		Likert order average	
Model	Pair acc.	AUC	Pair acc.	AUC
Gemma 2 2B	.680	.647	.627	.601
Gemma 2 9B	.907	.860	.947	.887
Qwen3 4B	.987	.991	1.000	.999
Qwen3 8B	1.000	.993	1.000	1.000
Llama 3.2 3B	.680	.624	.753	.669
Llama 3 8B	.633	.619	.680	.622

Table 2: Direct behavioral judgments on the same 300 held-out items. Scores are averaged within item across the Yes-or-No and No-or-Yes orders or across the ascending and reversed Likert anchors before computing the threshold-free pair-accuracy and ROC-AUC measures.

explanation, the continuation *and I don't know why* becomes reliably less expected even though the model is never directly asked whether the factor is causal.

The explicit-happenstance diagnostic is positive in all seven models as well. The critical-minus-control advantage for placing the same factor after *because* rather than *and it so happened that* ranges from 0.280 to 0.454 mean token log-probability units. Pairwise accuracy ranges from 0.760 to 0.940 and ROC-AUC from 0.750 to 0.907. Every pair-bootstrap interval excludes zero; all one-sided sign-flip tests attain the Monte Carlo floor ($p = 10^{-4}$). This result specifically concerns the explicit happenstance continuation, not a comparison with bare *and*.

These two diagnostics converge at the model level, but they test different consequences of an inferred explanation: incompatibility with professed ignorance and relative expectation under causal versus explicitly incidental framing. Their consistent direction establishes an output-level eliciture signature before the prompted and representational analyses below.

Direct prompted judgments provide a more mixed picture (Table 2). Qwen3 4B and 8B discriminate the conditions strongly under both endpoints: their order-averaged Yes/No ROC-AUC is 0.991 and 0.993, and their Likert ROC-AUC is 0.999 and 1.000, respectively. Gemma 2 9B retains useful rank discrimination (0.860 Yes/No and 0.887 Likert AUC). Gemma 2 2B and both Llama models are substantially weaker, with AUC from 0.601 to 0.669 and paired accuracy from 0.627 to 0.753 across endpoints.

The contrast between Tables 1 and 2 is therefore model-dependent rather than uniformly negative. Some models express causal sensitivity directly, but across the model set the more consistent behavioral signature appears indirectly in the likelihood of a downstream sentence: whether an

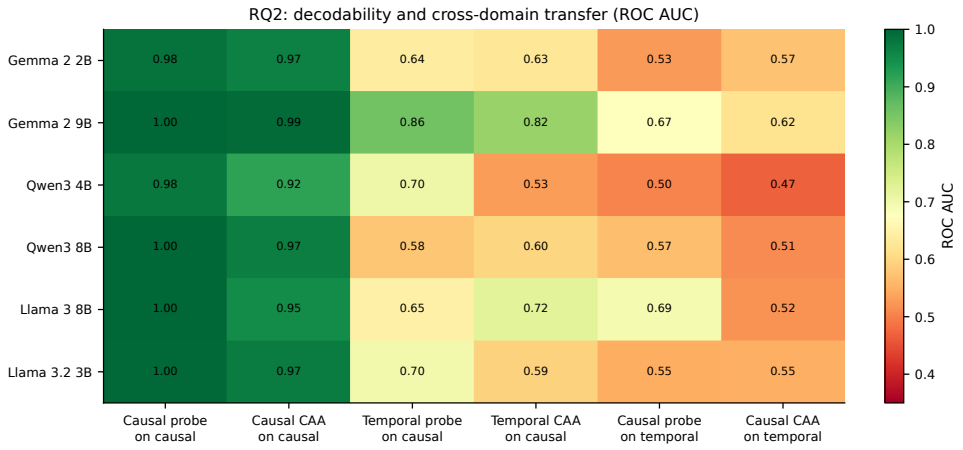


Figure 1: Development-selected probe and CAA ROC-AUC in domain and under causal–temporal transfer. All layers are frozen before test evaluation.

ignorance continuation is expected and whether a factor is favored under causal rather than happenstance framing. This dissociation suggests that implicit causal inference can be present even when a metalinguistic answer interface yields poorly calibrated or order-sensitive decisions.

3.2 RQ2: causal information is decodable, but temporal transfer is weak

Across the six models, causal probe ROC-AUC averages 0.992 and causal CAA ROC-AUC averages 0.961. Every causal probe exceeds 0.97 AUC, and paired accuracy is at least 0.987. The result is therefore consistent across model families and is not specific to a learned probe: a single mean-difference direction also discriminates held-out causal items well.

By contrast, temporal-to-causal transfer averages 0.686 AUC for probes and 0.647 for CAA. Figure 1 shows that transfer varies substantially by model and direction; several transferred CAA scores approach chance. Causal-to-temporal transfer is similarly inconsistent. Temporal precedence is therefore not a reliable linear substitute for graded causal implication, despite the high in-domain causal scores.

High representational discrimination does not guarantee calibrated behavior. Table 2 shows strong direct rank discrimination for Qwen3 4B and 8B but much weaker discrimination for Gemma 2 2B, Llama 3 8B, and Llama 3.2 3B. Fixed-threshold analyses additionally place the latter three near chance in balanced accuracy. Some of these failures are decision-level collapses: Gemma 2 2B predicts positive on 99.7% of items under Yes-or-No but on 0% under No-or-Yes, while its AUC remains 0.608 and 0.654, respectively. Llama 3 8B likewise shifts from a 93.7% to an 11.7% positive rate when answer order is reversed. Across models, the mean within-model balanced-accuracy range is 0.071 for Yes/No order and 0.046 for Likert anchor order. The rank-based metrics and fixed-threshold metrics thus diagnose different properties: encoded discrimination versus prompt-dependent decision calibration.

3.3 RQ3: representations track external ratings more consistently than self-ratings

Table 3 reports primary row-level Pearson correlations. Probe correlations with a model's own order-averaged behavior are high for Qwen3 4B and 8B (0.785–0.905), moderate for Gemma 2 9B (0.572–0.593), and weak for Gemma 2 2B and both Llama models (0.139–0.262). CAA follows the same broad pattern but is usually weaker than the probe.

In contrast, probe–Gemini correlations range from 0.796 to 0.922, including 0.829 for Gemma 2 2B, 0.912 for Llama 3 8B, and 0.871 for Llama 3.2 3B. These models therefore contain a linearly accessible ordering that agrees with the external graded judgment even when their own final prompted decisions are poorly calibrated. This dissociation argues against interpreting low behavioral balanced accuracy as absence of causal information.

Model	Probe–Y/N	Probe–Likert	Probe–Gemini	CAA–Y/N	CAA–Likert	CAA–Gemini
Gemma 2 2B	$.199 \pm .091$	$.149 \pm .094$	$.829 \pm .029$	$.181 \pm .100$	$.104 \pm .103$	$.785 \pm .032$
Gemma 2 9B	$.572 \pm .067$	$.593 \pm .057$	$.922 \pm .016$	$.548 \pm .076$	$.567 \pm .069$	$.840 \pm .026$
Qwen3 4B	$.785 \pm .034$	$.806 \pm .031$	$.796 \pm .036$	$.656 \pm .054$	$.712 \pm .040$	$.692 \pm .044$
Qwen3 8B	$.879 \pm .022$	$.905 \pm .015$	$.889 \pm .022$	$.765 \pm .034$	$.793 \pm .034$	$.774 \pm .041$
Llama 3 8B	$.139 \pm .081$	$.204 \pm .082$	$.912 \pm .014$	$.056 \pm .091$	$.244 \pm .089$	$.761 \pm .038$
Llama 3.2 3B	$.239 \pm .080$	$.262 \pm .077$	$.871 \pm .024$	$.245 \pm .087$	$.219 \pm .093$	$.798 \pm .031$

Table 3: Row-level Pearson correlations, shown as estimate $\pm$ half-width of the pair-clustered 95% bootstrap interval. Y/N and Likert targets are item-level order averages.

Model	Layer	BAcc	Acc.	AUC	Supp.	Post-hoc	Correct
Gemma 2 2B	17	.90	.89	.96	.92	.93	.86
Gemma 2 9B	26	.78	.85	.93	.88	.60	.84
Qwen3 4B	28	.68	.78	.84	.73	.49	.81
Qwen3 8B	34	.50	.92	.86	.14	.39	.96
Llama 3.2 3B	13	.84	.83	.93	.89	.86	.78
Llama 3 8B	17	.90	.88	.96	.95	.90	.84

Table 4: Frozen three-way probe-relative failure-probe performance on held-out items. Accuracy uses the probe's three-class argmax prediction, without a threshold. AUC is macro one-vs-rest ROC-AUC; the rightmost three columns are class recalls.

Spearman correlations reproduce the row-level pattern. Pair-difference correlations are considerably smaller: the largest probe Pearson value is 0.38 for Qwen3 4B and 8B, while most Gemini pair-difference correlations are near zero. The strong row-level association is therefore driven mainly by between-item ordering, not by covariation in the magnitude of each critical-minus-control shift. This distinction limits the claim: representations align with graded item plausibility, but do not robustly predict fine-grained paired effect sizes.

3.4 RQ4: probe-relative failure prediction transfers unevenly

The prompt-conditioned failure probe is evaluated on 1,200 held-out observations per model. Five of the six models transfer above chance, with balanced accuracies from 0.680 to 0.903 (Table 4). Gemma 2 2B and Llama 3 8B are strongest at 0.903 and 0.897 balanced accuracy, respectively, and attain macro one-vs-rest AUC above 0.96. Qwen3 8B is the exception: its high overall accuracy (0.917) reflects the dominant correct class, while its balanced accuracy is 0.497 and its post-hoc recall is 0.390. Thus, high causal discrimination in RQ2 does not guarantee that a model's decision failures occupy a stable, linearly transferable state.

3.5 RQ4: targeted interventions improve some endpoints

Classifier-gated steering has strongly model- and endpoint-dependent effects (Figure 2). The largest Yes/No AUC gains occur for Gemma 2 9B, from 0.830 to 0.985 ($\Delta = 0.155$), and Llama 3 8B, from 0.589 to 0.712 ($\Delta = 0.123$). Gemma 2 9B Likert improves from 0.866 to 0.919, and Llama 3.2 3B Likert improves from 0.636 to 0.710. The Qwen models show smaller positive changes. Effects need not improve every endpoint: Llama 3 8B Likert falls from 0.567 to 0.473, while Gemma 2 2B changes little.

Development selection retains $\alpha = 0$ when no positive penalized effect is supported, and some endpoint–failure combinations provide no estimable contrast vector. These null selections explain the unchanged aggregate cells. They are informative outcomes under this protocol, not evidence that no nonlinear or more localized intervention could work. More generally, subtracting a failure contrast and observing a predicted output shift provides causal evidence that the selected residual component influences the answer, but does not establish that the vector isolates the full mechanism of causal implication.

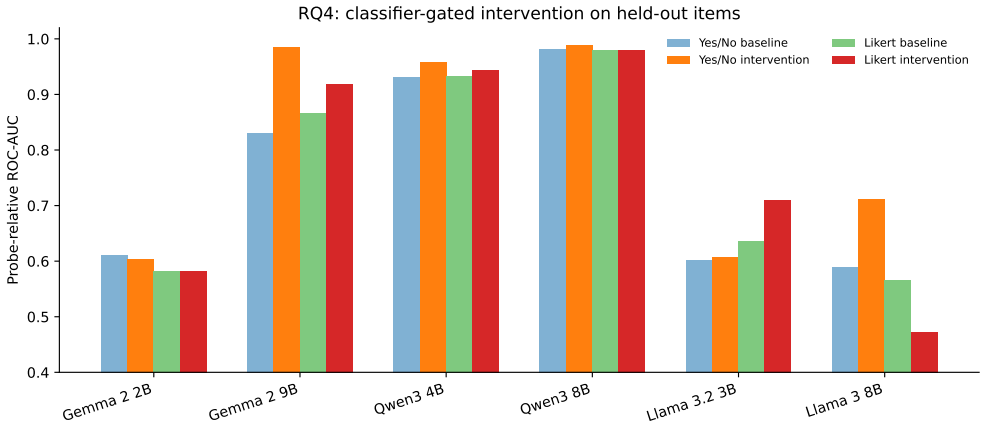


Figure 2: Baseline and classifier-gated intervention ROC-AUC against the frozen causal-probe reference on held-out items. Values average the two prompt orders within each endpoint.

	Model output	
Causal-probe decision	**Positive**	**Negative**
Positive	**True positive** Probe–output agreement	**Suppression** Probe positive, output negative
Negative	**Post-hoc attribution** Probe negative, output positive	**True negative** Probe–output agreement

Figure 3: Visual definition of the probe-relative RQ4 categories. *True positive* and *true negative* denote agreement with the frozen causal-probe decision; they do not assert agreement with dataset ground truth.

Limitations

Linear decodability does not identify a unique causal mechanism. The intervention vectors contrast naturally occurring error and matched-correct states and can therefore contain correlated features unrelated to the targeted failure. Prompt-order effects also show that failure labels partly depend on the measurement interface.

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A Probe-relative failure categories

Figure 3 summarizes the probe-relative labels used in RQ4. The frozen causal probe supplies the reference decision by thresholding its raw logit at zero. The prompted model decision is obtained from either the normalized Yes/No probability or normalized probabilistic Likert score at threshold 0.5; ties are negative. Labels are assigned separately to every item–prompt observation.

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